5: Part 1 - Data Visualization Basics

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Objectives

1. Perform simple data visualizations in the R package `ggplot`
2. Develop skills to adjust aesthetics and layers in graphs
3. Apply a decision tree framework for appropriate graphing methods

Opening discussion

Effective data visualization depends on purposeful choices about graph types. The ideal graph type depends on the type of data and the message the visualizer desires to communicate. The best visualizations are clear and simple. A good resource for data visualization is Data to Viz, which includes both a decision tree for visualization types and explanation pages for each type of data, including links to R resources to create them. Take a few minutes to explore this website.

Set Up

```
library(tidyverse);library(lubridate);library(here)
library(ggridges)

here()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
PeterPaul.chem.nutrients <-
  read.csv(here("Data/Processed_KEY/NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv"), string
PeterPaul.chem.nutrients.gathered <-
  read.csv(here("Data/Processed_KEY/NTL-LTER_Lake_Nutrients_PeterPaulGathered_Processed.csv"), stringsA
EPAair <- read.csv(here("Data/Processed_KEY/EPAair_O3_PM25_NC1819_Processed.csv"), stringsAsFactors = T

EPAair$Date <- ymd(EPAair$Date)
PeterPaul.chem.nutrients$sampledate <- ymd(PeterPaul.chem.nutrients$sampledate)
PeterPaul.chem.nutrients.gathered$sampledate <- ymd(PeterPaul.chem.nutrients.gathered$sampledate)
```

ggplot

ggplot, called from the package `ggplot2`, is a graphing and image generation tool in R. This package is part of tidyverse. While base R has graphing capabilities, ggplot has the capacity for a wider range and more sophisticated options for graphing. ggplot has only a few rules:

- The first line of ggplot code always starts with `ggplot()`
- A data frame must be specified within the `ggplot()` function. Additional datasets can be specified in subsequent layers.
- Aesthetics must be specified, most commonly x and y variables but including others. Aesthetics can be specified in the `ggplot()` function or in subsequent layers.
- Additional layers must be specified to fill the plot.

Geoms

Here are some commonly used layers for plotting in ggplot:

- geom_bar
- geom_histogram
- geom_freqpoly
- geom_boxplot
- geom_violin
- geom_dotplot
- geom_density_ridges
- geom_point
- geom_errorbar
- geom_smooth
- geom_line
- geom_area
- geom_abline (plus geom_hline and geom_vline)
- geom_text

Aesthetics

Here are some commonly used aesthetic types that can be manipulated in ggplot:

- color
- fill
- shape
- size
- transparency

Plotting continuous variables over time: Scatterplot and Line Plot

```
# Scatterplot ^ code above changes the layout of the chart/plot
ggplot(EPAair, aes(x = Date, y = Ozone)) +
  geom_point()
```

```
#create an object that stores the plot, define all the inputs, and then print plot
#this is useful for printing two plots in one window
O3plot <- ggplot(EPAair) +
  geom_point(aes(x = Date, y = Ozone)) #specify aesthetics
print(O3plot)
```

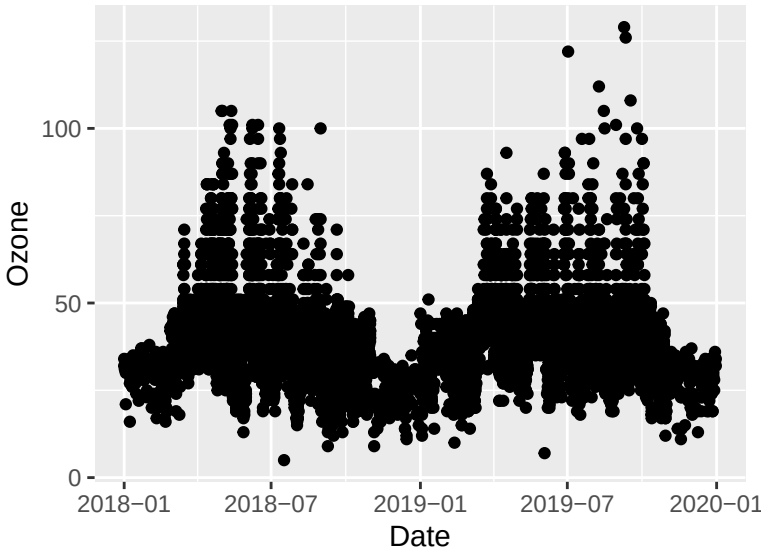


Figure 1: Figure 1

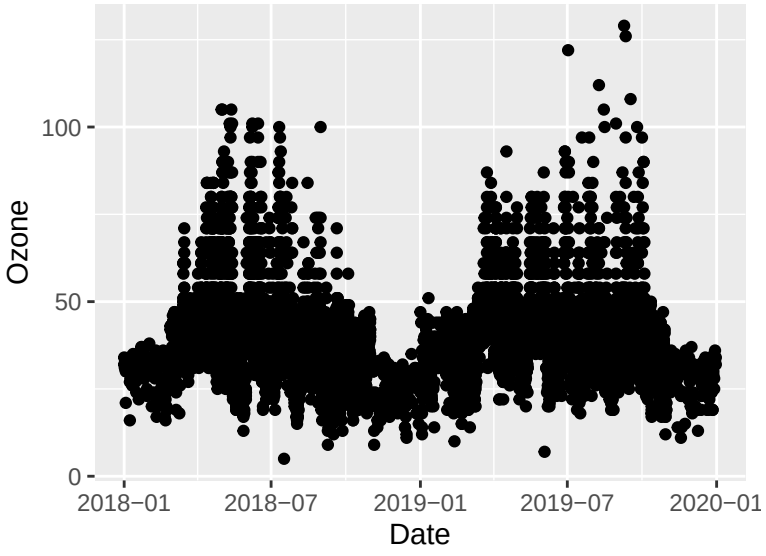


Figure 2: Figure 1

```r
# Fix this code
O3plot2 <- ggplot(EPAair) +
  # wrong -> geom_point(aes(x = Date, y = Ozone, color = "blue"))
  #change color outside the aes
  geom_point(aes(x = Date, y = Ozone), color = "blue")
print(O3plot2)
```

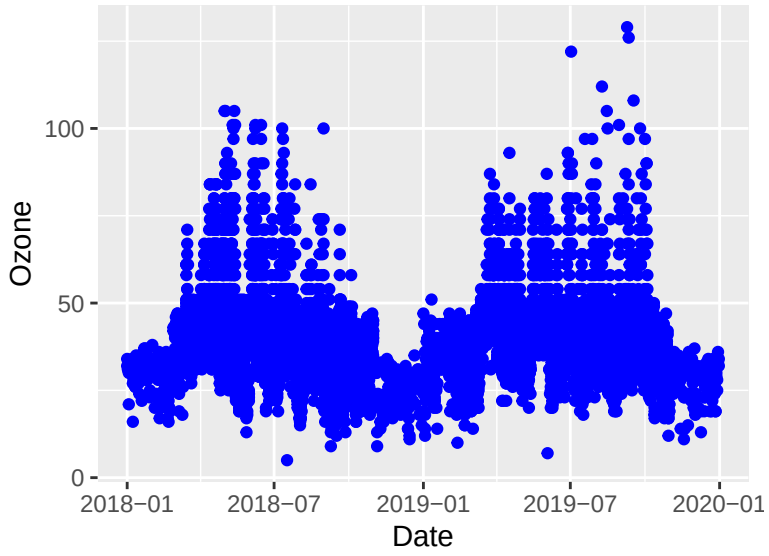


Figure 3: Figure 1

```r
# Add additional variables
# How could you automatically assign a marker color to a variable?
#shape of the dots = shape -> 2 diff based on year
#color code = color -> by site name
PMplot <-
  ggplot(EPAair, aes(x = Month, y = PM2.5, shape = as.factor(Year), color = Site.Name)) +
  geom_point()
print(PMplot)
```

```r
# Separate plot with facets, using .faceted
#
PMplot.faceted <-
  ggplot(EPAair, aes(x = Month, y = PM2.5, shape = as.factor(Year))) +
  geom_point() +
  facet_wrap(vars(Site.Name), nrow = 3) #instead of coloring sites w/ different colors, create a differ
print(PMplot.faceted)
```

```r
# Filter dataset within plot building and facet by multiple variables
#provide dataframe, use subset function to select sites
PMplot.faceted2 <-
  ggplot(subset(EPAair, Site.Name == "Clemmons Middle" | Site.Name == "Leggett" |
                  Site.Name == "Bryson City"),
```

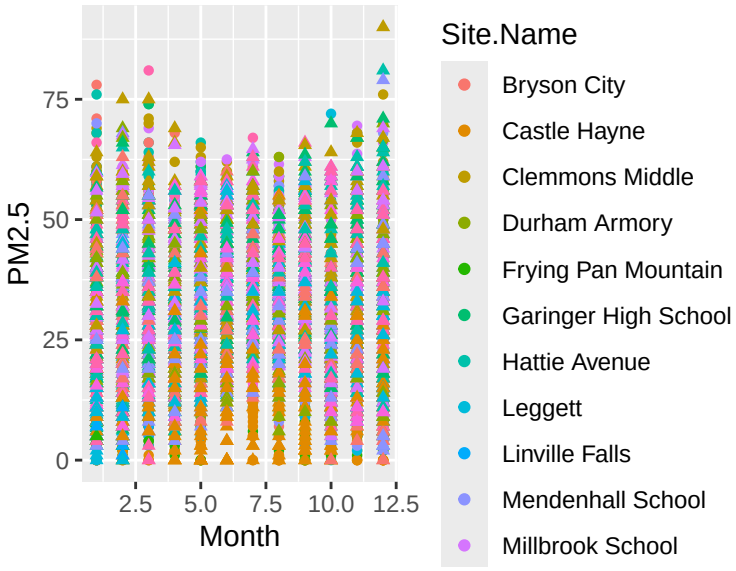


Figure 4: Figure 1

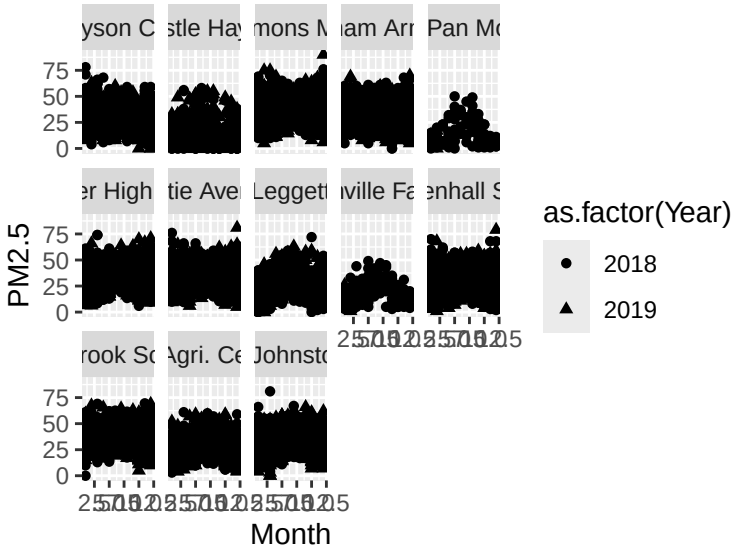


Figure 5: Figure 1

```
        aes(x = Month, y = PM2.5)) +
  geom_point() +
  facet_grid(Site.Name ~ Year) #site name by year, column year and row represent sites (grid like)
print(PMplot.faceted2)
```

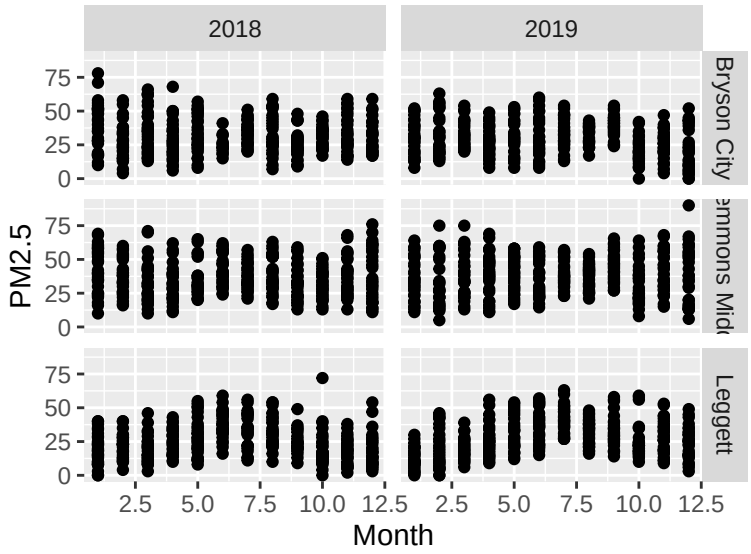


Figure 6: Figure 1

```
# Plot true time series with geom_line
PMplot.line <-
  ggplot(subset(EPAair, Site.Name == "Leggett"),
         aes(x = Date, y = PM2.5)) +
  geom_line()
print(PMplot.line)
```

Plotting the relationship between two continuous variables: Scatterplot

```
# Scatterplot
lightvsDO <-
  ggplot(PeterPaul.chem.nutrients, aes(x = irradianceWater, y = dissolvedOxygen)) +
  geom_point()
print(lightvsDO)
```

```
# Adjust axes
lightvsDOfixed <-
  ggplot(PeterPaul.chem.nutrients, aes(x = irradianceWater, y = dissolvedOxygen)) +
  geom_point() +
  xlim(0, 250) +
  ylim(0, 20)
print(lightvsDOfixed)
```

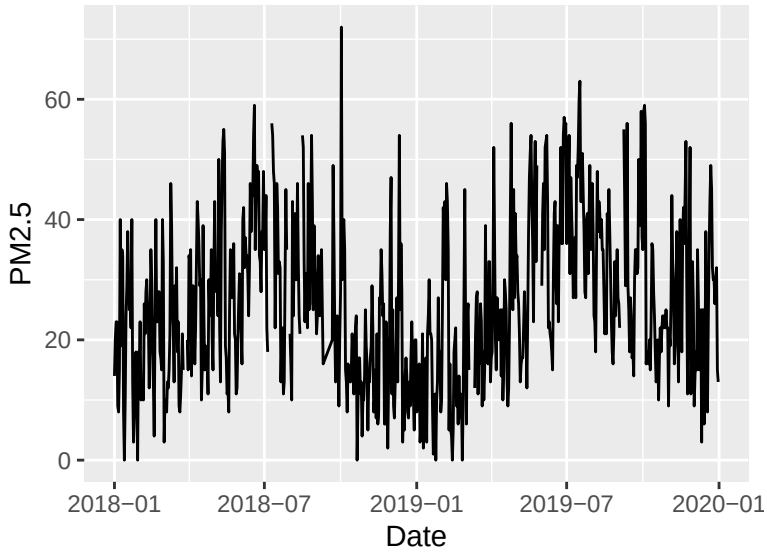


Figure 7: Figure 1

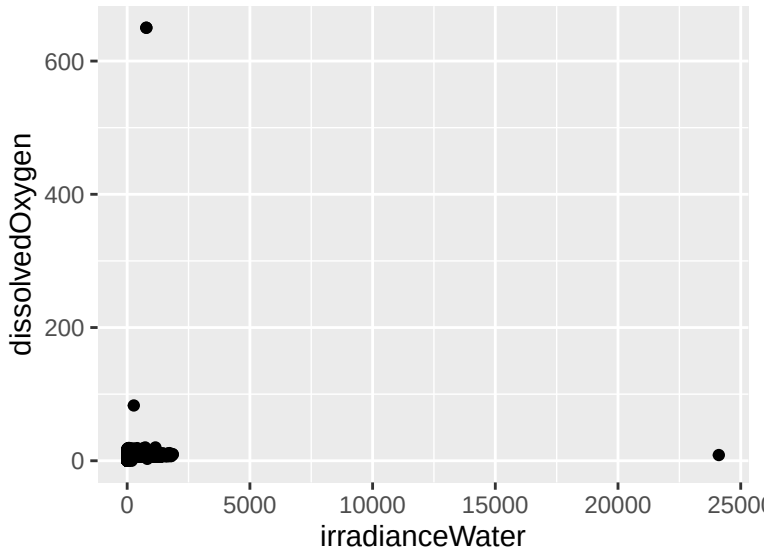


Figure 8: Another scatter plot

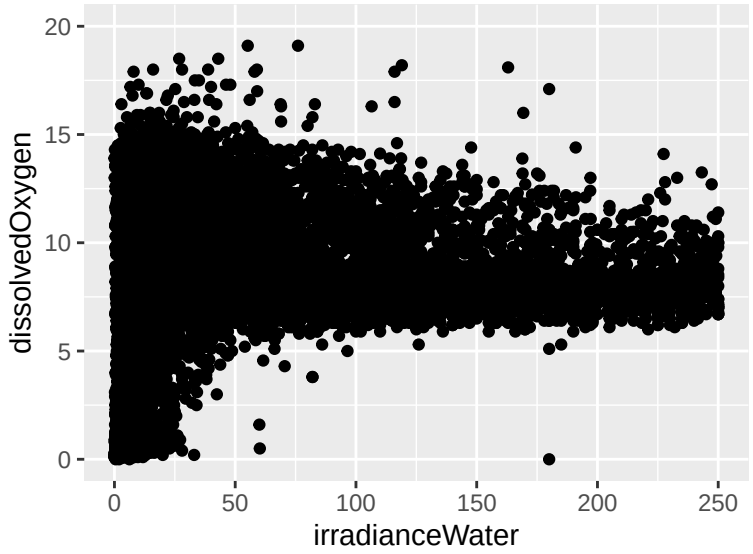


Figure 9: Another scatter plot

```
# Depth in the fields of limnology and oceanography is on a reverse scale
tempvsdepth <-
  ggplot(PeterPaul.chem.nutrients, aes(x = temperature_C, y = depth)) +
  #ggplot(PeterPaul.chem.nutrients, aes(x = temperature_C, y = depth, color = daynum)) +
  geom_point() +
  scale_y_reverse() #reveres y to have depth at the top
print(tempvsdepth)
```

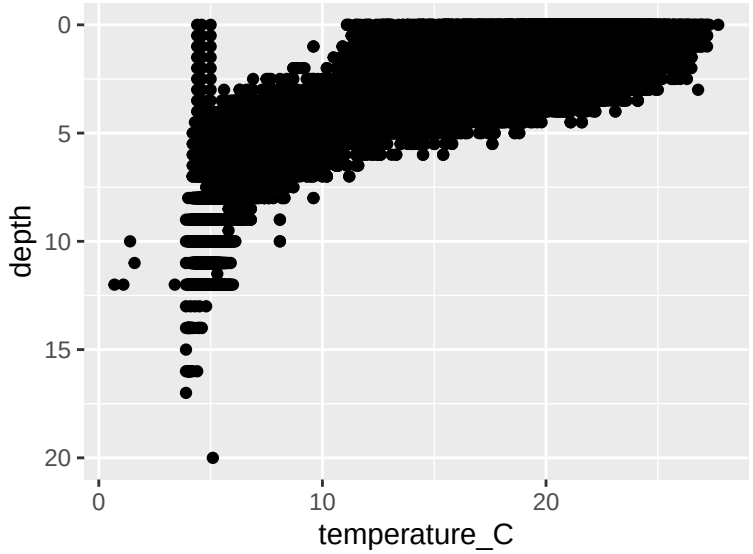


Figure 10: Another scatter plot

```r
#finding a trendline, geom_smooth (method = linear (lm))
NvsP <-
  ggplot(PeterPaul.chem.nutrients, aes(x = tp_ug, y = tn_ug, color = depth)) +
  geom_point() +
  geom_smooth(method = lm) +
  geom_abline(aes(slope = 16, intercept = 0))
print(NvsP)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

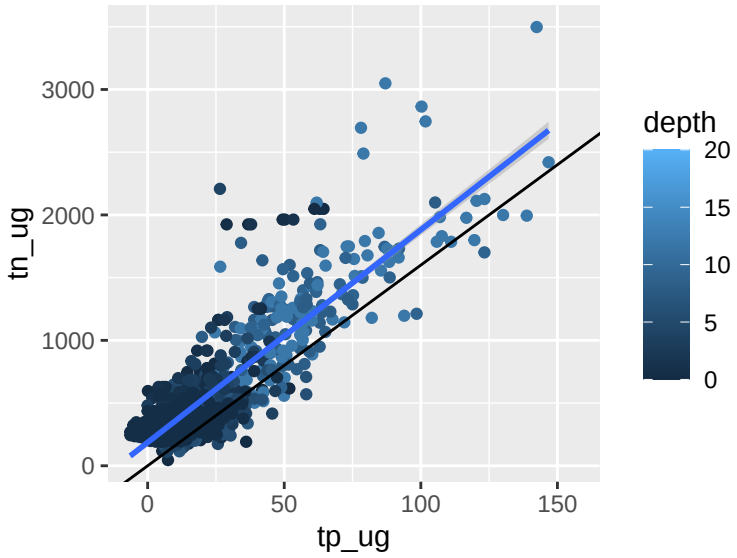


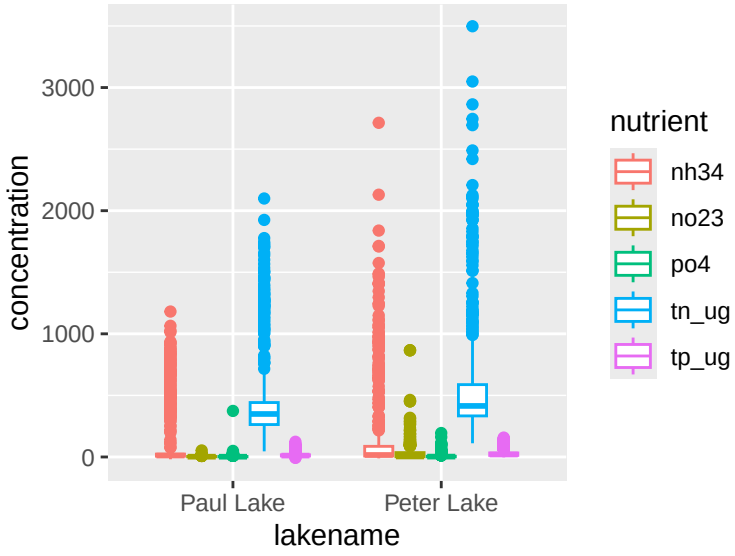
Figure 11: Another scatter plot

Plotting continuous vs. categorical variables

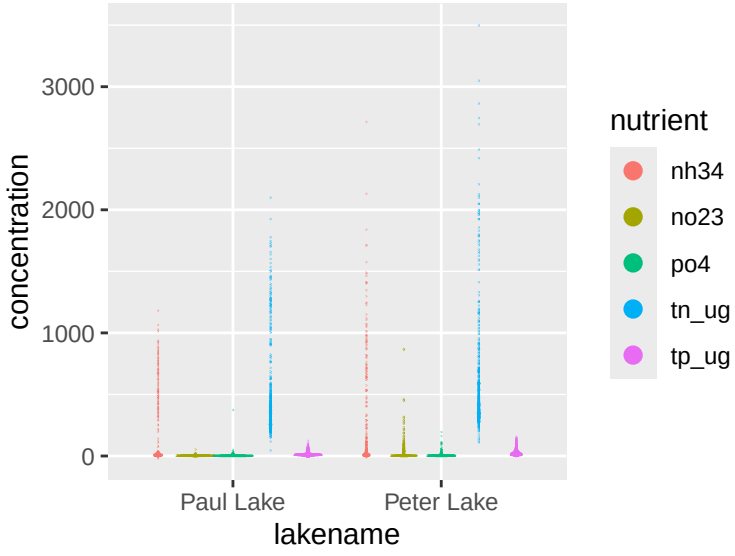
A traditional way to display summary statistics of continuous variables is a bar plot with error bars. Let's explore why this might not be the most effective way to display this type of data. Navigate to the Caveats page on Data to Viz (https://www.data-to-viz.com/caveats.html) and find the page that explores barplots and error bars.

What might be more effective ways to display the information? Navigate to the boxplots page in the Caveats section to explore further.

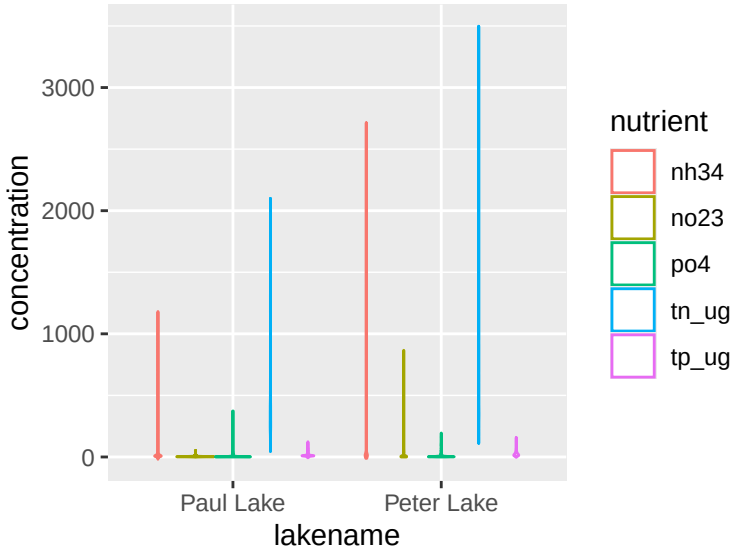
```r
# Box and whiskers plot
#gathered: 2 columns for the nutrients + concentration, shows all data
#separate color by nutrient column
#color vs. fill: color = contour of the shape, fill = fill of shape w/ black contours
Nutrientplot3 <-
  ggplot(PeterPaul.chem.nutrients.gathered, aes(x = lakename, y = concentration)) +
  geom_boxplot(aes(color = nutrient)) # Why didn't we use "fill"?
print(Nutrientplot3)
```




```
# Dot plot
#change y=axis -> binaxis = stacking dots along y-aixs, bin width= width of x-axis, position = moves do
Nutrientplot4 <-
  ggplot(PeterPaul.chem.nutrients.gathered, aes(x = lakename, y = concentration)) +
  geom_dotplot(aes(color = nutrient, fill = nutrient), binaxis = "y", binwidth = 1,
               stackdir = "center", position = "dodge", dotsize = 2) #
print(Nutrientplot4)
```




```
# Violin plot, similiar to boxplot (only show you summary stats), violin is a rotated density plot, doe
Nutrientplot5 <-
  ggplot(PeterPaul.chem.nutrients.gathered, aes(x = lakename, y = concentration)) +
  geom_violin(aes(color = nutrient)) #
print(Nutrientplot5)
```




```
# Frequency polygons
# Using a tidy dataset
#variable on x axis, and color of line
#no legend, no x-axis, not the best way to do this
Nutrientplot6 <-
  ggplot(PeterPaul.chem.nutrients) +
  geom_freqpoly(aes(x = tn_ug), color = "darkred") +
  geom_freqpoly(aes(x = tp_ug), color = "darkblue") +
  geom_freqpoly(aes(x = nh34), color = "blue") +
  geom_freqpoly(aes(x = no23), color = "royalblue") +
  geom_freqpoly(aes(x = po4), color = "red")
print(Nutrientplot6)
```

```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
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```
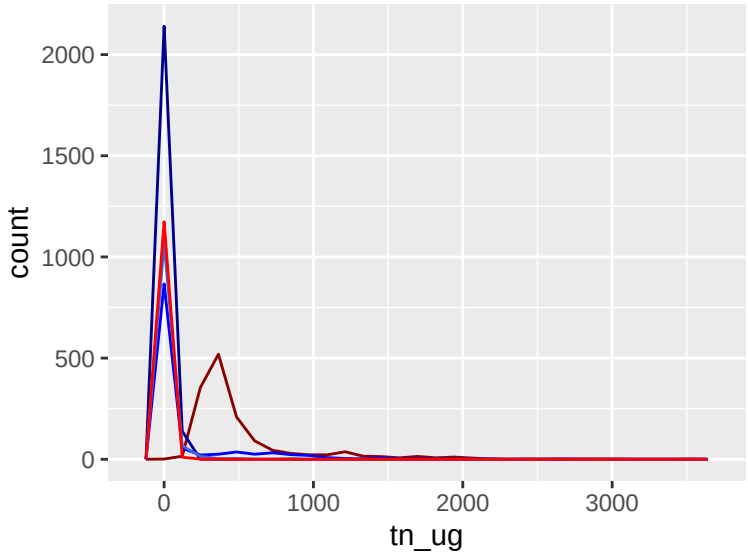
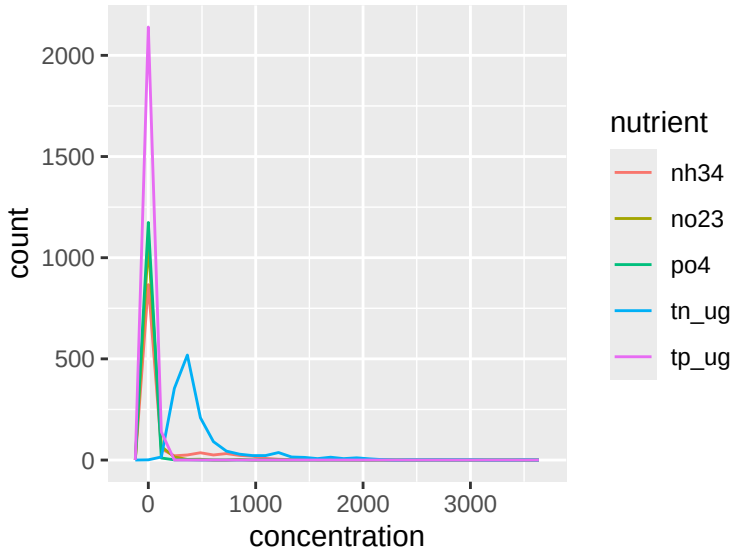




```
# Instead best doing with a gathered dataset, value of pivot wider/longer
#same frequency plot, nutrient gathered in one column and concentration in one, add one geom only
Nutrientplot7 <-
  ggplot(PeterPaul.chem.nutrients.gathered) +
  geom_freqpoly(aes(x = concentration, color = nutrient))
print(Nutrientplot7)
```

```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```




```
# Frequency polygons have the risk of becoming spaghetti plots.
# See <https://www.data-to-viz.com/caveat/spaghetti.html> for more info.

# Ridgeline plot
#y by nutrient, one ridge plot for each late
Nutrientplot6 <-
```

```
  ggplot(PeterPaul.chem.nutrients.gathered, aes(y = nutrient, x = concentration)) +
  geom_density_ridges(aes(fill = lakename), alpha = 0.5) #
print(Nutrientplot6)
```

```
## Picking joint bandwidth of 10.9
```